

Yuwen Fan (Wendy)

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WORKING & EDUCATION

Postdoc Associate in Hong Kong University of Science and Technology <i>Division of Environment and Sustainability</i>	2025-2026
Ph.D. in Hong Kong University of Science and Technology (4.3 / 4.3) <i>Doctor of philosophy in Atmospheric Environmental Science</i> (conferred on 15 th Oct 2025) Main Courses: Remote Sensing, Machine Learning	2022-2025
M.Phil in Hong Kong University of Science and Technology (4.15 / 4.3) <i>Master of philosophy in Atmospheric Environmental Science</i> Main Courses: Atmospheric Dynamics, Environmental Data Analysis	2020-2022
B.Sc. in Hong Kong University of Science and Technology (3.62 / 4.3 First Class) <i>Bachelor in Computer Science (Double major)</i> Main Courses: Machine Learning, Java Programming, Software Engineering, Database <i>Bachelor in Environmental Management and Technology (Double major)</i> Main Courses: Ecology, Biology, Environmental Science, GIS, ESG	2016-2020

PUBLICATIONS

Shen, H.¹, **Fan, Y.**^{1*}, Hur, J., and Im, E.-S.* (2026). Irrigation Impact on Humid Heat Stress: Divergent Metrics, Vegetation Dynamics, and Future Worsening. *Geophysical Research Letters*, under review.

Fan, Y.*, Pokhrel, Y., Shen, H., Hur, J., and Im, E.-S.* (2026). Decoding Convergent Hydroclimatic Responses to Irrigation in Structurally Distinct Regional Climate Models. *Journal of Climate*, in revision.

Fan, Y.* & Im, E.-S. (2025) The Role of Vegetation Dynamics in Assessing Irrigation Impacts. *Geophysical Research Letters*, 52(12), e2024GL114566. <https://dx.doi.org/10.1029/2024GL114566>

Fan, Y.*, Yang, Z., Lo, M.-H., Hur, J. & Im, E.-S.* (2025). Deciphering the Capricious Precipitation Response: Irrigation Impact in the North China Plain. *NPJ Climate and Atmospheric Science*, 8(1), 211. <https://doi.org/10.1038/s41612-025-01063-3>

Fan, Y.*, Yang, Z., Lo, M.-H., Hur, J. & Im, E.-S.* (2024). Applying double-cropping and interactive irrigation in the North China Plain using WRF4.5. *Geoscientific Model Development*, 17(18), 6929–6947. <https://doi.org/10.5194/gmd-17-6929-2024>

Fan, Y., Im, E.-S.*, Lan, C.-W., & Lo, M.-H.* (2023). An increase in precipitation driven by irrigation over the North China Plain based on RegCM and WRF simulations. *Journal of Hydrometeorology*, 24(6), 1155–1173. <https://doi.org/10.1175/JHM-D-22-0131.1>

Fan, Y.¹, Liao, P.-S.¹, Im, E.-S.*, & Lo, M.-H. (2022). Regional disparities in the exposure to heat-related mortality risk under 1.5 °C and 2 °C global warming. *Environmental Research Letters*, 17(5), 054009. <https://doi.org/10.1088/1748-9326/ac5adf>

Fan, Y. & Im, E.-S.* (2022). A Comparative Assessment of Changes in Heat-Related Mortality Risk Under the RCP2.6 and RCP8.5 Scenarios Based on the CORDEX-CORE Ensembles. *Asia-Pacific Journal of Atmospheric Sciences*, 59(2), 207–218. <https://doi.org/10.1007/s13143-022-00304-2>

(* means corresponding authors, 1 means the co-first author)

PREVIOUS PROJECTS

Improve the **crop system** in land surface models and quantify **irrigation impact** in North China Plain

(Related work has been published in JHM, GMD, npjCAS and GRL)

- Improve the irrigated-crop simulation including the implementation of double-cropping rotation, interactive irrigation, and groundwater pumping in WRF-NoahMP, RegCM-CLM, CESM-CLM.
- Validate the improved model's performance in simulating crop system (vegetation growth, irrigation amount, groundwater usage) and local climate (regional meteorology and hydrology).
- Quantify the irrigation impact on climate using three distinct model to improve the robustness.
- Identify the added-value of better land representation in assessing the irrigation impact, and highlight the potential deviations that could be made using the non-improved model.

Assess the **heat-related mortality risk** under global warming

(Related work has been published in ERL and APJAS)

- Calculating Excessive Mortality Risk and Environmental Health Factor to estimate the heat mortality.
- Understand the role of mitigation strategies in reducing heat-related mortality by comparing the indices under different future climate scenarios (i.e., 1.5°C & 2°C, RCP2.6 & RCP8.5)
- Identify the regional hotspots in terms of exposure of (the elderly) population to heat-related mortality risk, and explain the regional discrepancies.
- Estimate the uncertainty that could be raised from adopting different equations during the calculation.

EXPERIENCES & AWARDS

[Academic Service] Independent **Reviewer** in CEE, JGR-Atmosphere, JHM, ACCRE

[TA in PG course] Atmospheric Dynamics in Fall 2021, Fall 2022, Fall 2023, Fall 2024

[TA in UG course] Environmental Geographical Information System in Spring 2023

[TA in UG course] Climate Change Impacts and Extreme Weather Events in Spring 2021

2026 Houston Oral presentation in **AMS** and got **Best Student Oral Presentation** prize

2025 Trieste Oral lecture in **ICTP** workshop

2025 Singapore Oral presentation in **AOGS**

2025 Hong Kong AIS **Best Research** Award

2025 Hong Kong HKUST Redbird Award

2024 Vienna Poster presentation in **EGU**

2024 D.C. Poster presentation in **AGU**

2024 Michigan **Visiting scholar** in Michigan State University for six months

2024 Hong Kong HKUST Redbird Award

2023 Colorado Oral presentation in **AMS**

2022 Singapore Oral presentation in **AOGS**

2022 Hong Kong HKUST Redbird Recruitment Award

2020 Hong Kong University's Scholarship for students with top 10% academic performance

2019 Denmark Undergraduate exchange in Technical University of Denmark

2019 Spring RICS (Royal Institution of Chartered Surveyors) Scholarship

2018 Korea Summer exchange in Korea University

2017 Dalian Neusoft internship, duty in holding weekly teleconference with Amazon

TECHNICAL SKILLS

Programming: Python, Fortran, NCL, Matlab, C/C++, Java

Languages: Proficient in English (including academic presenting), native in Chinese

Knowledge : A+ in all major courses (e.g., Atmospheric Dynamic, Environment Data Analysis)

AI Background: SVM, FNN, CNN, RNN, LSTM

RESEARCH INTERESTS

Land Surface Modeling, Earth System, Land-Atmospheric Interactions, Regional Climate Modeling